

Tidy data

KMK

Tidy data

“Happy families are all alike; every unhappy family is unhappy in its own way.” — Leo Tolstoy

Tidy datasets are all alike but every messy dataset is messy in its own way.

Tidy data: A standardized way to link structure of a dataset with its semantics

- 80% of data analysis is spent on cleaning and preparing data

Tidy data: A standardized way to link structure of a dataset with its semantics

- Data structure
 - made up of rows and columns
 - columns almost always labelled

Table 1

person	treatmenta	treatmentb
John Smith	–	2
Jane Doe	16	11
Mary Johnson	3	1

Table 2

treatment	John.Smith	Jane.Doe	Mary.Johnson
treatmenta	–	16	3
treatmentb	2	11	1

Tidy data: A standardized way to link structure of a dataset with its semantics

- Data semantics
 - every value belongs to a *variable* and an *observation*
 - A variable contains all values that measure the same underlying attribute across units
 - An observation contains all values measured on the same unit across attributes

Table 3

name	treatment	result
John Smith	a	–
Jane Doe	a	16
Mary Johnson	a	3
John Smith	b	2
Jane Doe	b	11
Mary Johnson	b	1

Tidy data: A standardized way to link structure of a dataset with its semantics

Data semantics

it is easier to describe functional relationship between variables

it is easier to make comparison between group of comparison

id	sex	weight	height
1	Male	65	168
2	Female	55	170
3	Female	38	168
4	Male	44	172

Tidy data

1. Each variable forms a column.
2. Each observation forms a row.
3. Each type of observational unit forms a table.

name	treatment	result
John Smith	a	—
Jane Doe	a	16
Mary Johnson	a	3
John Smith	b	2
Jane Doe	b	11
Mary Johnson	b	1

Common problems of messy datasets

- Column headers are values not variable names.
- Multiple variables are stored in one column.
- Variables are stored in both rows and columns.
- Multiple types of observational units are stored in the same table.
- A single observational unit is stored in multiple tables.

Use melting, string splitting, and casting

Column headers are values not variable names

religion	<10k	10-20k	20-30k	30-40k	40-50k	50-75k
Agnostic	27	34	60	81	76	137
Atheist	12	27	37	52	35	70
Buddhist	27	21	30	34	33	58
Catholist	418	617	732	670	638	1116

Melt the columns into rows

religion	income	frequency
Agnostic	<10k	27
Agonistic	10-20k	34
Agonistic	20-30k	60

Multiple variables are stored in one column

country	year	m014	m1524	m2534	f014
AD	2000	0	0	1	—
AE	2000	2	4	4	3
AF	2000	52	228	183	93

First, melt the data

country	year	column	cases
AD	2000	m014	0
AD	2000	m1524	10
AD	2000	m2534	12
AE	2000	f014	3

Then, split the string column

country	year	sex	age	cases
AD	2000	m	0-14	0
AD	2000	m	15-24	10
AD	2000	m	25-34	12
AE	2000	f	0-14	3

Variables are stored in both rows and columns

id	year	month	element	d1	d2	d3	d4	d5
MX17004	2010	1	tmax	-	-	-	-	-
MX17004	2010	1	tmin	-	-	-	-	-
MX17004	2010	2	tmax	-	27.3	24.1	-	-
MX17004	2010	2	tmin	-	14.4	14.4	-	-
MX17004	2010	3	tmax	-	-	-	-	32.1
MX17004	2010	3	tmin	-	-	-	-	14.2

First melt the data

id	year	month	element	day	value
MX17004	2010	01	tmax	d30	27.8
MX17004	2010	01	tmin	d30	14.5
MX17004	2010	02	tmax	d2	27.3
MX17004	2010	02	tmin	d2	14.4
MX17004	2010	02	tmax	d3	24.1
MX17004	2010	02	tmin	d3	14.4

Create date using year, month and day, and drop missing values

id	date	element	value
MX17004	2010-01-30	tmax	27.8
MX17004	2010-01-30	tmin	14.5
MX17004	2010-02-02	tmax	27.3
MX17004	2010-02-02	tmin	14.4
MX17004	2010-02-03	tmax	24.1
MX17004	2010-02-03	tmin	14.4

Next cast tmax and tmin

id	date	tmax	tmin
MX17004	2010-01-30	27.8	14.5
MX17004	2010-01-30	27.3	14.4
MX17004	2010-02-02	24.1	14.4

Multiple types of observational units are stored in the same table.

- e.g.,
 - volunteers/doctors and patients
 - clinic/hospital ward and patients
 - dental patients and teeth

Song,artist and rank in one table

year	artist	time	track	date	week	rank
2000	2 Pac	4:22	Baby Don't Cry	2000-02-26	1	87
2000	2 Pac	4:22	Baby Don't Cry	200-03-04	2	82
2000	2Ge+her	3:15	The Hardest Part of...	2000-09-02	1	91
2000	2Ge+her	3:15	The Hardest Part of...	2000-09-09	2	87
2000	3 Doors Down	3:53	Kryptonite	2000-04-08	1	81
2000	3 Doors Down	3:53	Kryptonite	2000-04-15	2	70

Song, artist and rank in one table

- duplication with artists and times repeated
- break down into
 - artist, song name and time
 - rank of song and week
- Relational data
- Also require merging the two data set in one table using unique id

Song, artist and time

id	artist	track	time
1	2 Pac	Baby Don't Cry	4:22
2	2Ge+her	The Hardest part of ...	3:15
3	3 Doors Down	Kryptonite	3:53
4	3 Doors Down	Loser	4:24

Dates and rank of song

id	date	rank
1	2000-02-26	87
1	2000-03-04	82
1	2000-03-11	72
1	2000-03-18	77
1	2000-03-25	87
1	2000-04-01	94
1	2000-04-08	99
2	2000-09-02	91
2	2000-09-09	87
2	2000-09-16	92
3	2000-04-08	81
3	2000-04-15	70
3	2000-04-22	68
3	2000-04-29	67
3	2000-05-06	66

One type in multiple table

- e.g.,
 - participants of survey 1 and 2
 - data of three quarterly reports
- Add a new column that records the original table
- Then append the tables

Survey 1 table

id	age	sex	Knowledge score
FHW01	34	M	70
FHW02	30	F	76
FHW03	38	F	69

Survey 2 table

id	age	sex	Knowledge score
MPS01	39	M	78
MPS02	40	M	74
MPS03	38	F	70

Appended

id	age	sex	Knowledge score	survey
FHW01	34	M	70	1
FHW02	30	F	76	1
FHW03	38	F	69	1
MPS01	39	M	78	2
MPS02	40	M	74	2
MPS03	38	F	70	2